A Structural Model of Automated Market Making

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November 18, 2025†

CBER CtCe Working Paper Series‡

Abstract

Automated market makers (AMMs) process billions in annual transactions, yet most rely on fixed fee schedules that stand in contrast to microstructure theory, which prescribes volatility-sensitive spreads. To assess whether this theory extends to AMMs, we develop and estimate a structural model of an AMM. We show that fixed fees are inefficient and characterize the optimal volatility-sensitive fee schedule. Testing on ETH-USDC data shows that, even with noisy volatility forecasts, adaptive fees outperform fixed fees, increasing the annual fee revenue by 9-44%, and AMM liquidity supply by 2-10%.

Keywords: Automated Market Makers (AMM), Decentralized Finance (DeFi), Dynamic Fees, Market Microstructure, Uniswap, Volatility.

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†First version: October 3, 2023.

‡The authors are grateful to the Uniswap Foundation for funding provided through the CBER CtCe grant.

1 Introduction

Decentralized finance is reshaping trade by enabling markets that function without centralized intermediaries. Automated Market Makers (AMMs) such as Uniswap have cumulatively processed trillions of dollars in transactions, suggesting that algorithmic market making can operate at institutional scale. In response, a growing literature has begun to examine how AMMs might reshape market structure. For example, Malinova and Park (2023) asks whether AMMs could meaningfully complement, or even eventually replace the traditional limit-order markets.

Despite their success, AMMs make design choices that depart from core principles of traditional market microstructure, particularly in how they manage liquidity provider (LP) risk. A core tenet of traditional centralized exchanges is that market makers widen spreads when information asymmetry or volatility rises, which protects them from adverse selection and inventory risk. Leading AMMs, in contrast, typically rely on fixed fee percentages that do not adjust with market conditions. AMMs' fixed-fee approach represents either a costly oversight or evidence that their trading mechanism can achieve near-optimal outcomes with a simple constant-fee policy. This is the question we address in this paper: is the current fee structure, with a constant fee level, optimal? Or, to put it differently, could an adaptive, volatility-sensitive fee policy lead to more efficient AMM performance?

We find that volatility-sensitive fees generate substantial efficiency gains. Using a structural model calibrated to transaction-level Uniswap data, we find that volatility-sensitive fees yield 13.2% higher annual fee revenue (range 9% to 44%) and attract 2.6% more liquidity (range 2% to 10%) relative to fixed fees. The optimal fees we derive increase with expected volatility, which is intuitive because elevated price volatility has a similar effect in the AMM setting to elevated adverse selection under a traditional market design, e,g, a limit-order book.

AMMs differ fundamentally from traditional centralized exchanges, therefore it is unclear in principle whether the absence of adaptive fees is of first-order economic importance. In centralized limit-order markets, prices emerge from the continuous interaction of buy and sell orders that traders submit, cancel, and revise. By contrast, AMMs rely on a transparent, pre-specified pricing curve such as a constant product that maps the pool's asset balances into a quoted price, allowing trades to execute directly against pooled liquidity rather than against standing orders. Liquidity providers (LPs) supply capital to these pools and earn fees on each trade. As trading activity changes the pool's asset balances, the AMM moves mechanically along its pricing curve, generating fee revenue.

Despite these architectural differences, AMMs expose liquidity providers to economic forces that parallel those faced by market makers in traditional limit-order markets. The mechanisms differ, but

the risks that determine compensation for liquidity provision are closely aligned. Liquidity providers are exposed to "impermanent loss," defined as the shortfall in value relative to a chosen benchmark. Common benchmarks include a passive buy-and-hold or buy-and-stake position in the underlying assets, or an actively and continuously rebalanced (hedged) strategy (Milionis et al. 2022). All yield the same economic conclusion: absent trading fees, LP returns systematically underperform these benchmarks in expectation. Such losses by LPs are analogous to market makers in centralized markets incurring losses to high-frequency traders taking advantage of stale limit orders. AMMs generate a similar effect: as external prices change, arbitrageurs trade against the pool to realign its quoted price, mechanically rebalancing LP portfolios toward the asset that has depreciated and away from the one that has appreciated. This mechanism operates even under full information. A simple example illustrates the effect. Suppose an AMM pool initially holds 10 ETH and \$1,000 USDC, implying a price of \$100 per ETH with constant product $k = 10{,}000$. If the external price doubles to \$200, arbitrageurs buy ETH from the pool until its price matches, yielding new reserves $x_1 = \sqrt{50} \approx 7.07$ ETH and $y_1 \approx 1{,}414$ USDC. Ignoring fees from these trades, the LP's position is worth about \$2,828 versus \$3,000 under buy-and-hold, highlighting the inventory distortion created by the AMM's pricing rule.

A second, related parallel is adverse selection. In limit-order markets, market makers lose when facing better-informed traders. AMMs admit a similar channel: traders who observe new price information before it is incorporated into the pool can transact against stale AMM prices and earn rents at the expense of LPs. This mechanism is distinct from impermanent loss.

A third parallel involves order-processing and latency costs. In centralized markets, these costs arise from updating quotes, monitoring inventory, and processing trades in real time. AMMs exhibit analogous frictions due to blockchain settlement: LPs can adjust positions only discretely, face confirmation delays and transaction costs, and cannot respond instantaneously to price changes. These constraints limit real-time inventory management and mirror the operational frictions embedded in traditional spreads. Taken together, these parallels indicate that AMMs reproduce the fundamental economic forces that shape compensation for liquidity provision, even though their trading protocols differ from those in centralized markets.

Microstructure literature (Kyle 1985, Glosten and Milgrom 1985, Easley and O'Hara 1987) shows that costs of liquidity provision, e.g., due to information asymmetry, are reflected in trading frictions (bid-ask spread or market impact) faced by the market participants. Against this backdrop, the fixed-fee structures used by AMMs stand out as a gap in their design.

One potential reason why existing AMMs use constant fee schedules is legacy – early AMM

designs did not accommodate dynamic fee schedules. Another potential reason is the challenge of empirical identification of the optimal fee schedule. In AMMs, fees, liquidity, and trading volume are ideally jointly determined in equilibrium. LPs choose how much capital to supply based on expected fees and impermanent-loss risk, while traders respond to both fee levels and available liquidity. These simultaneous decisions create a deep endogeneity problem: changes in fees affect liquidity and volume, but liquidity and volume also affect the fees that would be optimal. As a result, the causal effect of fee changes on market outcomes cannot be recovered with standard empirical approaches or simple reduced-form variation. Previous theoretical work, including Evans et al. (2021), analyzes optimal AMM fees in stylized settings but does not address the empirical identification of fee effects in environments where fees, liquidity, and volume are jointly determined.

With recent technological innovations, the idea of adaptive fees has become increasingly visible in practitioner discussions. Shin and Adams (2023) argues that in a very volatile asset you might need higher fees," and Ippolito (2023) similarly notes that one of the big topics in DEX-land right now is dynamic fees." These views suggest that fixed fees may leave value on the table for both LPs and exchanges. Our conversations with researchers at Uniswap indicate that this perspective has gained additional momentum with the launch of Uniswap v4 in 2025. The v4 architecture introduces "hooks," customizable smart-contract modules that allow pools to adjust key aspects of trade execution, including fee logic, making state-contingent fees operational at scale. Consequently, dynamic fees have become a prominent topic of discussion in decentralized exchange design, even though their economic implications remain largely unexplored.

We address these gaps by developing a structural framework that captures the equilibrium relationships among fees, trading volume, and liquidity provision. The model jointly characterizes the behavior of LPs, who allocate capital based on expected returns and impermanent-loss risk, and traders, whose order flow reflects both noise trading and arbitrage activity. Estimating these decisions in a unified system allows us to recover the causal effect of fee policy in an environment where fees, liquidity, and volume are determined simultaneously. The framework delivers closed-form expressions for optimal fees as functions of expected volatility and market conditions, providing a rigorous basis for evaluating the performance of adaptive fee schedules.

To quantify the magnitudes involved, we apply the framework to Uniswap v2 transaction data for the ETH–USDC pair. The available AMM data contains no fee variation, making it impossible to estimate fee elasticities directly. Our structural approach addresses this constraint by tracing out the range of elasticities consistent with observed behavior and evaluating AMM performance across these plausible values. We combine external price data to forecast volatility with a predictive model

of impermanent loss and use these components to simulate daily optimal fees under alternative elasticity assumptions. Across all specifications, the gains from volatility-sensitive fees are robust to different trader responses to fee changes. These counterfactuals speak directly to the design choices now feasible in protocols such as Uniswap v4.

In summary, the paper makes four contributions. First, it develops a formal framework that embeds AMM fee-setting within market microstructure economics, clarifying how volatility, inventory risk, and trader behavior jointly determine optimal fees. Second, it directly tackles the core endogeneity problem in AMMs – fees, trading volume, and liquidity provision being jointly determined – by introducing a structural empirical approach that identifies these equilibrium relationships in the absence of fee variation. To our knowledge, this is the first paper to address this identification challenge directly. Third, it provides the first systematic evidence on how AMM liquidity and volume respond to fee changes by estimating these components with on-chain transaction data. Finally, it quantifies the performance of volatility-sensitive fees using counterfactual simulations grounded in the estimated model, showing their potential to substantially improve AMM performance, and the trillions of dollars in transactions that this mechanism supports.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 presents the theoretical framework and outlines the three-shot game. Section 4 derives the equilibrium. Section 5 describes the data and the two-stage GMM estimation. Section 6 outlines the in-model impact of a dynamic fee. Section 7 concludes.

2 Literature

Although we are unaware of any prior structural estimation work that focuses primarily on optimal AMM *fee structure*, several papers have developed theoretical models that capture different aspects of the dynamics of AMMs. Several of these introduce theoretical models of AMMs, but they miss certain crucial ingredients necessary to model how liquidity providers and traders react to changes in fees.

In order to capture these dynamics we need strategic liquidity providers who react to (1) the volatility of the asset price (which can cause impermanent loss) and (2) trader volume. Then we also need traders who react to (1) the amount of liquidity in the pool (this determines price slippage) and (2) the trading fees.

Below, we outline how prior works model liquidity providers and traders, and highlight how they are insufficient to comprehensively address the problem of optimizing fees.

One work which considers how fees affects liquidity provider profits is Evans et al. (2021). In their setting, LPs are not interested in maximizing returns, but instead on keeping their portfolio close to some optimal portfolio weights (e.g. 50% ETH 50% USDC). With this objective, higher fees mean that there is a large "no-arb window" where arbitrageurs will not adjust the pool weights. This leads them to the rather counterintuitive conclusion that "liquidity provider's losses are minimized by reducing the fees as much as possible, while ensuring they are not zero." Their model does not formally address other trading behavior.

Four works that introduce (and solve) models for strategic liquidity provisioning are Lehar and Parlour (2021), Hasbrouck et al. (2022), Capponi and Jia (2021), Malinova and Park (2023).

Lehar and Parlour (2021) introduces a one-step model. LPs are strategic, and decide how much liquidity to put into the pool, then with some fixed probability, there is a price shock. If there is a price shock, the pool is arbitraged back. If there is no price shock, there is a noise trader, then an arbitrage trade which pushes the pool back to the true price. In their setting, the noise trader trades a fixed amount independent of the pool size and fees. This would imply that the fee could be set arbitrarily large without affecting noise trader demand. Thus this approach is not directly applicable to our setting.

Hasbrouck et al. (2022) introduces a multi-period model. LPs are strategic, and decide how much liquidity to put into the pool. Then, at every time step, noise traders with idiosyncratic demands arrive probabilistically, and trade with the pool. In this model, there are no price fluctuations of the assets, so the LPs are not subject to impermanent loss.[1] In our model, we must take into account price fluctuations, when considering setting the optimal fee.

Capponi and Jia (2021) is a two-step model in which LPs are strategic and deposit at time 0. Then at time 1, there is either a price shock or a noise trade. This is similar to Lehar and Parlour (2021), but the key difference is that the price shock / noise trade is not immediately arbitraged. Instead, at time 2, an arbitrageur arrives, and the LP and the arbitrageur engage in a gas auction, where the LP tries to withdraw (to avoid impermanent loss), and the arbitrageur tries to arbitrage the pool. They show, however, that the arbitrageur will *always* win this auction whenever there is more than one LP, because the arbitrageur will make gains proportional to the entire pool size, whereas the LP's savings are only proportional to their share of the pool. They identify when there is a "liquidity-freeze," i.e., when LPs deposit zero liquidity in equilibrium.

[1] "Crucially, it is well-established that fluctuations in the fair value exchange rate between cryptoassets in a pool generate losses, termed impermanent loss (see Capponi and Jia 2021), that then necessitate fees to compensate liquidity providers. We emphasize that the channel we establish does not depend on such price fluctuations and we hold constant the fair value exchange rate to establish that point."

Malinova and Park (2023) introduces a multi-step model and attempts to determine if AMMs would be a viable mechanism for traditional assets. Fee is strategically set to minimize transaction costs for traders and then liquidity providers deposit according to a break-even constraint. They empirically test their model using equities data and find AMMs can significantly reduce trader costs. However, in this model, while traders split their flow between an AMM and a traditional market based on their costs, the total volume is exogenous and fixed.

Adams et al. (2024) Considers an alternative, "auction-managed" AMM (am-AMM) model, where individuals can purchase the rights to "manage" the AMM. The manager pays the LP's for this privilege, and in turn is able to set AMM fees, and collects *all* fees on trades. This effectively allows the manager to trade without fee, and thus the manager is the only arbitrageur who can (profitably) trade in the no-arb window. They model noise traders whose trade volume depends on both fees and liquidity.

Milionis et al. (2022) introduces the concept of Loss Versus Rebalancing (LVR). In their model, liquidity is fixed (LPs are non-strategic), and they do not specify any model for noise traders or fees[2]. In Milionis et al. (2023), this model is extended to account for fees paid by *arbitrageurs*, but the model does not include noise traders. In both these works, liquidity is fixed, so there is no attempt to solve for equilibrium levels of liquidity.

Angeris et al. (2021) creates a multi-period *simulation-based* analysis of AMM trading and liquidity provision. Their model includes fees and noise traders[3]. In their model, they discuss how increases in fees increase the "no-arbitrage" window, but they do not explore how changes in fees affect liquidity provider profits.

Rao and Shah (2023) proposes a new fee mechanism, where fees are not constant in trade size, but instead the marginal fee is decreasing, i.e., the total fee is *concave* in the trade size (as opposed to linear in trade size). Using these new "triangle fees," they simulate LP profits when the prices move according to symmetric random process, and arbitrageurs rebalance the pool. Their model assumes liquidity is fixed, they do not model noise traders.

In this work, we present the first structural AMM model that includes (1) strategic LPs, (2) traders that are sensitive to fees and liquidity, and (3) price shocks, and yet is simple enough that we can characterize the amount of liquidity in equilibrium and test it empirically.

[2]They define a variable, Fee_t, denoting the fees collected at time t, but they do not give it any specific form.

[3]Since noise traders are not strictly rational, it is necessary to make assumptions about how they trade. In Angeris et al. (2021), noise traders will trade a fixed amount with the AMM, if and only if the AMM price is no more than a fixed percentage away from the true price. This means that noise traders will not trade if the fees are too high, or liquidity is too low.

3 Model

Consider an exchange setting with three types of strategic agents: a platform operating an AMM, liquidity providers (LPs), and traders. Each make decisions/take actions in their own dedicated time periods, denoted by $t = 0$ (platform), $t = 1$ (LPs), and $t = 2$ (traders). At time $t = 0$, the platform sets the fees, f. At time 1, the LPs deposit an amount of capital, c. At time $t = 2$, traders trade with the platform, and there is a price shock. After trades have been made at time $t = 2$, all agents exit and the game ends. The timeline details are provided further down.

Some may note that we do not include arbitrageurs in our model explicitly. It is reasonable to consider an extension of our model where there is a fourth time period $t = 3$ where arbitrageurs arrive and trade the AMM to match the external price. However, as we discuss later on, this extension is already accounted for within our model and is effectively equivalent. Intuitively, this is because our model considers trading by both noise traders and arbitrage trades in aggregate, without needing to distinguish one from the other. The only assumption needed is that the AMM price is in line with the external price at the end of each period, which we discuss futher below and additionally show is reasonable in our empirical study.

The Financial Market

The AMM has a liquidity pool composed of two generic assets. To ease exposition, we refer to it as the "risky/stable" pool, e.g. ETH/USDC; the stable asset is the numeraire, and all quantities will be expressed in its units, referred to as \$ for short. Let p_0 be the fair market price of the risky asset on the AMM at time 0, before any trades. For instance, $p_0 \approx \$3,500$ in the case of the price of ETH at the time of writing. The risky asset's price has a random fundamental value at the end of the game, which we model as

$$p = v \cdot p_0,$$

where v is a random variable representing percent change, with PDF φ_v, mean μ and volatility σ. When considering an explicit form for φ_v, we use a lognormal distribution.

We consider a simple Constant Function Market Maker (CFMM) invariant with a proportional trading fee f (e.g., Uniswap v2). The CFMM exchange rules work as follows: for a pool with ℓ_s units of the stable asset, ℓ_r units of the risky asset, and pool fee f, an incoming trade swapping δ_s units of the stable asset for δ_r units of the risky asset (buying the risky asset) must satisfy

$$(\ell_s + (1 - f) \cdot \delta_s)(\ell_r - \delta_r) \geq \ell_s \cdot \ell_r \qquad \text{(Buying from the pool)} \tag{1}$$

The fee is always taken in the incoming asset with respect to the pool. That is, a sell of δ_r units of the risky asset to the pool must satisfy

$$(\ell_s - \delta_s)(\ell_r + (1-f)\cdot\delta_r) \geq \ell_s \cdot \ell_r \qquad \text{(Selling to the pool)} \tag{2}$$

A graphical depiction of this invariant or "bonding curve" can be seen in Figure 1.

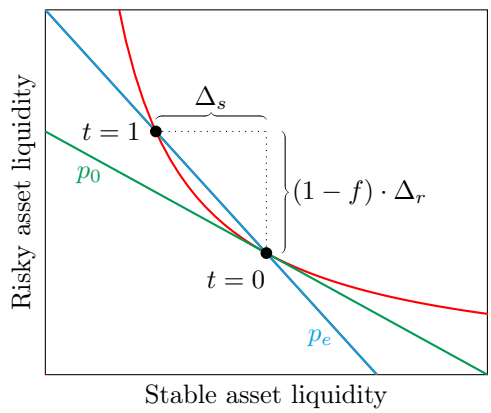

Figure 1: A visual illustration of a CFMM trade. The curve in red is the invariant. A trader is selling Δ_r units of the risky asset in return for Δ_s units of the stable asset. The effective price p_e is the (negative) slope of the curve shown in blue. The instantaneous price p_0 can then be determined as the slope of the tangent at the current liquidity amounts, that is, the derivative of the invariant. This is shown in green.

The Platform

We view the platform as an entity similar to the Uniswap Foundation. In this light, it is reasonable to assume that the platform wishes to set the fee f to maximize deposited liquidity c.

$$\max_f \quad c \tag{3}$$

The key consideration is that the fee level will affect the endogenous entry of LPs and the liquidity amount they provide. We describe their behavior next.

The Liquidity Providers

LPs deposit stable and risky amounts that are equivalent in value based on the current fair price of the pool. As total liquidity deposited c must correctly reflect the price p_0, this results in the initial constraints $\ell_s = p_0 \cdot \ell_r$ and $\ell_s + p_0 \cdot \ell_r = c$, which implies

$$\ell_s = \frac{c}{2} \quad \text{and} \quad \ell_r = \frac{c}{2 \cdot p_0}.$$

Thus, as initial price is exogenous, we can simplify the LPs' decision to a single variable c, rather than ℓ_r, ℓ_s.

LP revenue ultimately comes from a single source: fees from trading volume $F(\cdot)$. However, since the price at the end of the game is not necessarily the same as it was in the beginning, LPs also incur a profit or loss to their portfolio depending on if the final price increases or drops. We assume that the ending price of the period aligns with the external price. That is, the price of the AMM moves from p_0 to $v \cdot p_0$ where v is stochastic and represents the percentage movement of the price.

We additionally assume that the liquidity market is perfectly competitive, implying that LPs are strategic and provide enough liquidity to break even against a baseline portfolio that is similar to their pool reserve allocation. In practice, using Ethereum as an example, traders can stake their Ethereum directly by running a validator or through a service such as Lido. Additionally, they could make return on their USDC by providing liquidity on AAVE or simply moving it to Coinbase. To model this, instead of the usual buy-and-hold or loss-versus rebalancing benchmarks considered in the AMM literature, we consider a more stringent benchmark which we refer to as the "buy-and-stake" portfolio (BNS). The return on BNS over a time period dt is defined as

$$\frac{c}{2} \cdot e^{\gamma_s \cdot dt} + \frac{c}{2 \cdot p_0} \cdot e^{\gamma_r \cdot dt} \cdot v \cdot p_0 = c \cdot \left(\frac{e^{\gamma_s \cdot dt} + e^{\gamma_r \cdot dt} \cdot v}{2} \right) \tag{4}$$

where γ_s is the required return on the stable asset and γ_r is the required return on the risky asset. We notate the multiplier as $\gamma(v)$ for brevity, which we define as

$$\gamma(v) = \left(\frac{e^{\gamma_s \cdot dt} + e^{\gamma_r \cdot dt} \cdot v}{2} \right). \tag{5}$$

The portfolio value of an LPs' deposit after the pool is moved to a new price $p_0 \cdot v$ can be calculated through the CFMM invariant. Specifically, the reserve amounts of the LP become

$$\left(\frac{c \cdot \sqrt{v}}{2}, \frac{c}{2 \cdot p_0 \cdot \sqrt{v}} \right) \tag{6}$$

which, in terms of the stable asset, results in a final value of

$$c \cdot \sqrt{v}. \tag{7}$$

It is important to note that if one only considers the reserve amounts of the LP compared to even a simple "buy-and-hold" strategy, the LP will be guaranteed a loss. This is known as *impermanent*

loss in the literature and has been widely studied. Specifically, we see that

$$c \cdot \frac{1+v}{2} \geq c \cdot \sqrt{v} \tag{8}$$

holds directly from the AM-GM inequality. Thus, intuitively, it should be the case that fees from trading volume cover this deficit. However, given that trading volume is more volatile than a relatively risk-free staked portfolio, we additionally consider a "risk premium". Specifically, we postulate that since LPs are investing in a riskier asset, they may expect a higher return on average than the baseline.

LPs will deposit enough liquidity such that the returns of the AMM are equivalent to the outside option of BNS. The resulting break-even condition is given by

$$\mathbb{E}\left[F(\cdot) + c \cdot \sqrt{v}\right] = \mathbb{E}[c \cdot \gamma(v)] + c \cdot RP. \tag{9}$$

where RP is the discussed risk premium. We maintain as our baseline assumption that RP is a constant over time, which we empirically investigate later in section 5.2.2.[4]

Traders

We assume there is a volume V of trading that is dependent on liquidity c, fee f, and price volatility σ. This extends many existing noise trading models in the AMM literature, which often assume noise trading volume is exogenously fixed to a constant Malinova and Park (2023), Hasbrouck et al. (2023) (in Milionis et al. (2022), liquidity is fixed to a constant). However, determining the exact amount of fees earned by LPs from trading volume is complex as this not only depends on the size of each trade but also the *order* in which they occur. In light of this, we instead approximate trading fees as

$$F(\cdot) \approx f \cdot V(f, \sigma, c) \tag{10}$$

This approximation has been made in various other works such as Milionis et al. (2022), but its accuracy has not yet been empirically validated. We provide this validation in section 5.2.1, and generally it does extremely well.

We model expected trading demand to be linear in σ, f and $\log c$. We find that this form is quite accurate in our empirical investigation, but models that capture more nuanced behavior can

[4]It is straightforward to extend our model by allowing the conditional RP to depend on observables, e.g., the lagged liquidity level or price volatility. This would not change our analysis qualitatively, but would potentially require a longer time-series to estimate the risk-premium parameters with sufficient accuracy.

be used as well. For simplicity, we consider the following form for trading demand:

$$\mathbb{E}[V(f, \sigma, c)] = a_0 - a_f \cdot f + a_\sigma \cdot \sigma + a_c \cdot \log c \tag{11}$$

where structural parameters $a_f, a_\sigma, a_c > 0$.

We hypothesize that traders are inversely sensitive to trading fee f, positively sensitive to price volatility σ, and positively sensitive to liquidity c. The first two assumptions are natural, and the third can be thought of as sensitivity to price slippage, i.e. as liquidity increases, less slippage occurs so more trading happens. The choice of $\log c$ is determined by the assumption that traders are more sensitive to fee than they are to slippage (for large enough liquidity). Intuitively, as c approaches infinity, trading volume should not increase linearly since slippage becomes effectively zero. Otherwise, an LP would be able to generate infinite returns with infinite liquidity, an impractical result.

We acknowledge that our choice of function does not hold intuitively at edge conditions, i.e. trading volume should be 0 when fee is 1 or liquidity is 0. However, in our empirical study we show that this linear function is highly representative of the real world and should hold for small changes in parameters. In particular, trading volume increases slightly with liquidity and more drastically with external price volatility. It is important to note that since the historical data we have obtained is under a single fee used in Uniswap V2, it is not possible to directly estimate a_f. However, intuitively, it should be the case that as fee increases, trading volume decreases, as reflected by the negative sign in front of a_f. Estimation of this parameter would become feasible if and when AMMs eventually switch to more dynamic fee structures.

Arbitrage Assumption

We assume that arbitrage activities in our environment are unconstrained. As a result, in equilibrium, it should be impossible to earn abnormal risk-adjusted returns by trading on the differences between the price of the risky asset implied by the AMM and its external price. For example, prices implied by Uniswap consistently track traded prices on Binance Lehar and Parlour (2021). We operationalize this assumption by assuming that, at the end of each period, the AMM price of the risky asset equals its external price. Because arbitrage is subject to transaction costs in our model, the arbitrageur's optimal trade when the pool price and external price are misaligned should move the AMM price to the closest *edge* of the no-trade window. Thus, our assumption of convergence at the end of the period could potentially bias our estimation of the expected impermanent loss.

This could happen if the final pool price predictably diverged from $c\sqrt{v}$ on average, or if the initial spread between the pool price and the external price predicted the impermanent loss of the LPs. We discuss this in more detail in section section 5.2.3, where we find that neither type of predictability is supported by our empirical evidence.

Preferences and Beliefs

All agents are risk-neutral and know the distribution of the price process (v). They are able to view all past actions played, e.g. historical liquidity or trading volume.

Timeline

Figure 2 summarizes the timing of aforementioned events.

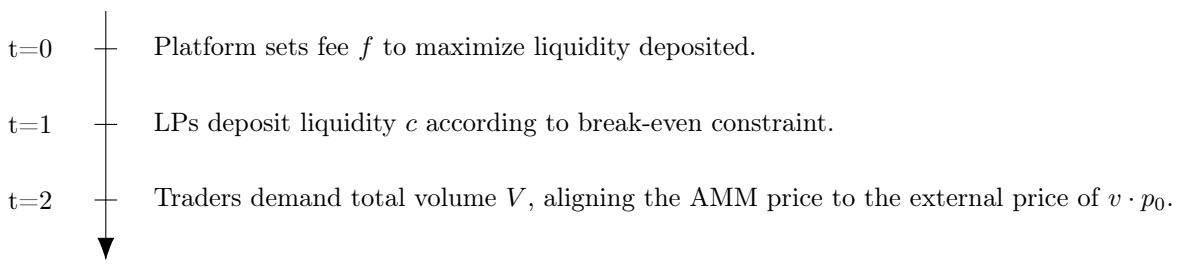

Figure 2: Model timeline.

4 Analysis

While our model is approximate, its simplicity allows us to derive closed forms. First, we show some preliminary results that will aide our subsequent results. All proofs can be found in the appendix.

Lemma 1 (BNS Impermanent Loss)**.** *If a liquidity provider makes no fees, then their portfolio will never be as good as the BNS portfolio. That is,*

$$\mathcal{L} = \gamma(\mathbb{E}[v]) - \mathbb{E}[\sqrt{v}] \geq 0 \tag{12}$$

Note that the liquidity provider's break-even constraint can be rewritten as

$$f \cdot V(f, \sigma, c) = c \cdot (\mathcal{L} + RP) \tag{13}$$

which leads to a very natural interpretation: fees from trading volume should cover impermanent loss with repsect to a buy-and-stake portfolio. For the rest of the paper, when we refer to the

break-even constraint, we consider this form.

Lemma 2 (Trade Volume Response to Optimal Fee)**.** *If liquidity and fee are set as best responses according to the break-even constraint, trading volume must respond negatively to increases in fee. In other words, it must be the case that*

$$\frac{\partial V}{\partial f} < 0 \tag{14}$$

Importantly, lemma 2 holds for any functional form of trading volume and leads to a natural result. That is, if an AMM follows the break-even constraint, traders must negatively react to increases in fee.

4.1 Static Fees

Building further on lemma 2 with our specific functional form for trading volume, the first-order condition of the break-even constraint with respect to the optimal fee gives us

$$a_0 - a_f \cdot f + a_\sigma \cdot \sigma + a_c \cdot \log c - a_f \cdot f = 0 \tag{15}$$

$$a_0 - a_f \cdot f + a_\sigma \cdot \sigma + a_c \cdot \log c = a_f \cdot f \tag{16}$$

Note that if a platform sets a static fee f, in order for it to be optimal, the above must be satisfied. This effectively states that if one uses a static fee, then trading volume should always stay *constant* in expectation. Particularly, changes in price volatility should not affect trading volume. However, a simple empirical test on real-world data correlating daily trade volume and price volatility shows that this is clearly not the case. Thus, we can immediately reject the hypothesis that a static fee is optimal.

Thus, from here, we reform our goal of determining if the current fee structure is optimal to instead exploring how an optimal fee should behave under the break-even constraint.

4.2 Simplified Model

Before we proceed to analyze the full model, we consider a simplified model that enables us to obtain interpretable closed-form solutions and tease out one of the underlying economic forces driving our main result. The key assumption we make in this section is that for some pools, particularly those with ample liquidity, trade demand may be relatively insensitive to small changes in liquidity

amounts, meaning $a_c \sim 0$. This leads to the linear approximation

$$V(c, f, \sigma) \approx V(f, \sigma) = a_0 - a_f \cdot f + a_\sigma \cdot \sigma \tag{17}$$

Solving the break-even equation for LPs (9) results in the following equilibrium liquidity injected by LPs

$$c^* = \frac{f \cdot (a_0 - a_f \cdot f + a_\sigma \cdot \sigma)}{\mathcal{L} + RP}. \tag{18}$$

This is concave in fee, we can therefore obtain the optimal fee by taking first-order conditions, leading to the formal result below.

Theorem 1 (Liquidity Insensitive Simplified Model)**.** *In the simplified model where traders are insensitive to liquidity, the optimal fee that maximizes liquidity c^* given by* (18) *is linearly increasing in the risky asset's return volatility, and is given by*

$$f^*(\sigma) = \frac{1}{2a_f}\Big(a_0 + a_\sigma \cdot \sigma\Big). \tag{19}$$

Theorem 1 provides a clear rationale for why fees should adapt to increases in returns volatility. But the only parameter driving this relationship in this simplified model is the a_σ term, which originates from the assumed functional form of the trading volume V. In other words, in situations where the trading volume may be entirely insensitive to volatility ($a_\sigma \sim 0$), then the optimal fee in (19) becomes constant! But this outcome is sensitive to the fact that the simplified model ignores endogeneity between trading volume and liquidity. Adding back endogeneity could reverse the result for the following reason: even if one assumes that $a_\sigma = 0$, volatility could still affect the optimal fee indirectly. This is because volatility affects impermanent loss (see (12)), which affects liquidity provisioning via the breakeven condition (13). In turn, the amount of liquidity injected into the pool would affect trading volume, which would then imply a possible relationship between fee and volatility, even as $a_\sigma \sim 0$.

To summarize, there are two possible forces at play that could justify dynamic fees as a function of volatility: The first is a direct relationship between trading volumes and volatilities ($a_\sigma > 0$). The second is an indirect relationship that occurs due to the endogenous link between trading volume and liquidity provisioning.

Adding this endogeneity substantially complicates the analysis. The rest of the paper is thus devoted to studying this more realistic situation.

4.3 Full Model with Endogeneity

Theorem 2 (Break-even Liquidity)**.** *Using the full model of volume eq.* (11)*, liquidity providers' best response to a fee f set by the platform is defined by*

$$c^* = -\frac{f \cdot a_c}{\mathcal{L} + RP} \cdot W\left(-\frac{\mathcal{L} + RP}{f \cdot a_c} \cdot \exp\left(\frac{a_f \cdot f - a_0 - a_\sigma \cdot \sigma}{a_c}\right)\right) \tag{20}$$

where $W(\cdot)$ is the Lambert W function.

Note that the interior of $W(\cdot)$ is always negative. Since the Lambert W function takes on two real solutions between $(-1/e, 0)$, if break-even liquidity has a solution, there must be two. Intuitively, the result of theorem 2 having two solutions comes from trader's sensitivity to liquidity. More complex forms of liquidity sensitivity can result in more solutions.

To classify each solution, consider that for any given liquidity amount in the pool, if LPs are making more than the outside option of BNS, it is strictly better for traders currently holding a BNS portfolio to instead put their assets into the pool. The reverse holds true if LPs are making less than the outside option: LPs would prefer to withdraw their holdings and reinvest into a BNS portfolio. This observation motivates the introduction of "stable" and "unstable" liquidity amounts, inspired from the traditional definitions in game theory.

Definition 1 (Stable Liquidity)**.** *A break-even liquidity c^* is **stable** if there exists a constant $\tau > 0$ such that for all $\epsilon \leq \tau$, $c^* + \epsilon$ results in less returns for liquidity providers than the outside option and $c^* - \epsilon$ results in more returns for liquidity providers than the outside option. That is, movements away from the break-even liquidty results in a pressure in the opposite direction. In other words, if LPs make less than BNS with liquidity greater than the break-even and more than BNS with liquidity less than the break-even, the break-even is stable.*

*An **unstable** break-even liquidity is where the opposite is true. That is, movements away from the break-even liquidity results in pressure to continue moving in that direction.*

Given this definition, we can now classify the two solutions found in theorem 2.

Theorem 3 (Stable Break-even)**.** *Of the two best responses the liquidity provider has, the lesser is unstable and the larger is stable. Thus, the break-even liquidity the provider prefers will be the larger solution.*

As a note, if one assumes that returns for LPs approach 0 as liquidity approaches inifinity, theorem 3 can be extended to generalize for a break-even condition with any number of solutions.

By simple contradiction, if the solution with the largest liquidity is unstable, then LPs must be making non-zero returns at infinite liquidity.

While it is technically possible to write a form for the platform's best response f^*, it appears to be inordinately large and complex, with no real analytical insights easily determined. Instead, we turn towards numerical simulations based on estimated parameters to determine behavior of optimal fee. Particularly, we aim to highlight the relationship between f^* and σ, as hinted at in the simple model's analysis. The next section is dedicated to this task.

5 Empirical Estimation

In this section we describe and implement our approach to estimating parameters of our structural model. In particular, we express the theoretical relation (11) as the following statistical relation:

$$V(f, \sigma, c)_t = a_0 - a_f f_{t-1} + a_\sigma \sigma_{t-1} + a_c \log c_{t-1} + \varepsilon_t \tag{21}$$

Here we assume that the process of the per-period realized price volatility is Markovian, and approximate the conditional expectation of the trading volume to be linearly related to the realized price volatility over the most recent period, σ_{t-1}. We further assume that the expected volume depends linearly on the end-of-period $t-1$ log liquidity level $\log c_{t-1}$, which is equivalently the beginning-of-period t log liquidity.

To estimate (21) empirically, we must address the challenge posed by the endogeneity of pooled liquidity. We assume that liquidity providers set their positions at the beginning of each trading period based on their information set, before observing the day's realized volatility and trading volume. As a result, we allow for the possibility that $\log c_{t-1}$ and ε_t are correlated, and we therefore cannot estimate the relation (21) by OLS. To address this, we use the Generalized Method of Moments (GMM). Instead of the conditional moment $\mathbb{E}_{t-1}[\log c_{t-1}\varepsilon_t]$, we impose the cross-equation restriction that the conditional mean of the risk-adjusted return spread between the AMM and the buy-and-stake strategy is zero. As we discuss below, we define several unconditional moments based on this conditional-moment condition.

Finally, note that parameter a_f cannot be identified based on available empirical information, because in Uniswap v2 only a single fee of 30 basis points has been used throughout our sample: $f_t = f$ is a constant. We therefore define a new parameter $\bar{a}_0 = a_0 - a_f f$ and re-write the volume equation as

$$V(f, \sigma, c)_t = \bar{a}_0 + a_\sigma \sigma_{t-1} + a_c \log c_{t-1} + \varepsilon_t \tag{22}$$

5.1 Data Collection

Uniswap Data

Data was collected by scraping all event logs from the Uniswap V2 ETH-USDC pool[5]. Logs were taken from the creation of the contract on May 5, 2020 to December 16, 2024. These event logs were then split based on their event type and parsed into CSV files. Specifically, Uniswap V2 pools emit 4 events of relevance: Mint, Burn, Swap, and Sync. A description of each of these events can be found in table 1. Ethereum event logs were specifically gathered from the publically available dataset on Google BigQuery Day and Medvedev (2018). Once downloaded, they were parsed into CSV formats. Only Swap and Sync events were collected and used in our analysis.

Event	**Description**	**Total**	**Post Merge**
Mint	LP deposits liquidity into the pool.	-	-
Burn	LP withdraws liquidity from the pool.	-	-
Swap	Swap is made against the pool.	5,840,505	2,003,132
Sync	Reserve amounts changed, i.e. burn/mint/swap/hotfixes.	5,918,265	2,012,786

Table 1: An overview of the event logs collected from the Uniswap V2 ETH-USDC pool. PM represents the total counts "post-merge", which enabled ETH staking and happened on September 15, 2022 at block 15537393.

Staked Returns Data

For the returns of staking both ETH and USDC, we used data gathered from Lido and AAVE. Lido is a DeFi pooling service where users can easily stake their ETH without the requirement of needing 32 ETH and hardware that is consistently connected to the Ethereum network. In short, trusted validators stake users' ETH for them in return for a small fee. AAVE is a DeFi lending service where users can provide liquidity to make passive return from lending. While the service is not completely risk free, as of writing, no lender has ever lost collateral on Aave.[6].

Data was gathered by using a full-archival Ethereum validator. We queried for the state of the Lido smart contract at a 100 block granularity, which translates to roughly a 20 minute window,

[5]Event logs were taken by scraping all logs emitted by address `0xb4e16d0168e52d35cacd2c6185b44281ec28c9dc`.

[6]Aave v2 had thousands of liquidations where *borrowers* have lost money, but the platform itself has only lost about $2M in "bad" loans. More than half of this loss was due to a single malicious loan taken out by Avi Eisenberg, the Mango markets attacker Warmuz et al. (2022). The losses to bad loans are miniscule compared to the total profits of the platform, and these losses have been absorbed by Aave itself, and no *lenders* have ever lost funds on Aave.

and the AAVE smart contract at 1000 block granularity, which translates to roughly 200 minutes. Full details of how the data was parsed can be found in our GitHub repository.

External Price Data

Since our model requires an assumption that the underlying asset pair can be traded externally to realize arbitrage opportunities, we turn towards Binance for price data. Binance is a centralized exchange where users can swap cryptocurrenies for each other and consistently has the largest trading volume out of all of the centralized exchanges. In particular, Binance has an ETH-USDC instrument for which we scraped price data at a 1 second granularity. Again, the code to scrape and merge price data files can be found in our GitHub repository.

5.2 Preliminary Results

We first empirically show some preliminary results that validate choices made in our model, particularly the choice of the trading fee approximation and the exclusion of explicitly modeling arbitrage trading.

5.2.1 Accuracy of Trading Fee Approximation

We first quantify the accuracy of approximating trading fees. In practice, fees are on the incoming asset, meaning on a buy of the risky asset, the fees are collected in the stable asset and on a sell of the risky asset, fees are collected in the risky asset. Thus, the direct calculation of fees earned from noise trades becomes fairly complex, since both the *size* and *ordering* of trades affect the total outcome. Simplifying assumptions such as perfectly balanced buys and sells do not appear to lessen the complexity either. Specifically, we would have that fees earned from a buy and sell of size x_n is

$$f \cdot \left(\frac{c \cdot p_0 \cdot x_n}{(1-f)(c - 2 \cdot p_0 \cdot x_n)} + x_n \cdot \mathbb{E}[v] \cdot p \right) \tag{23}$$

Instead, we consider the much simpler approximation of *expected* trading fees as

$$f \cdot V(c, f; v) \tag{24}$$

where V is the *expected* trading volume over the time period, which we posit is a function of liquidity, fee, and price volatility.

This approximation is motivated by the fact that one side of fees collected, i.e. in the stable

asset, is always simply f times the amount of the incoming asset. Thus, this approximation is exactly correct roughly half the time.

We quantify the accuracy of this approximation by comparing the difference between the actual fees collected for each day on the Uniswap V2 ETH-USDC pool from a month after its creation (May 2020) to August 20, 2024 (1542 days). The first month includes various test trades that are inane, resulting in poor representation of our estimate. For each day, the total volume is used to determine our fee approximation. To compute actual fees, we take only the inputs to the AMM and use the closet price from Binance to convert ETH into USD. Overall, we find that for practically all days, the fee approximation has less than 1% difference from the actual fees collected. Only 11 days have a larger than 1% difference with the largest being about 7.2%. A graph of the distribution of the differences can be seen in fig. 3.

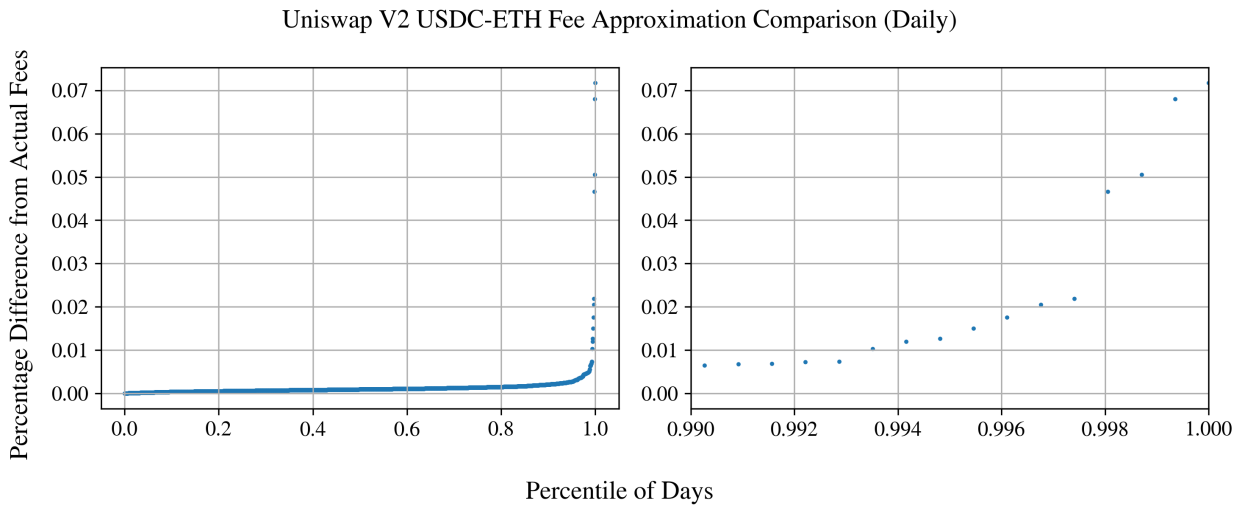

Figure 3: The distribution of the accuracy of the linear fee approximation method. Actual and approximate fees made were calculated for each day from June 1, 2020 to August 20, 2024 (1542 days). The percentage difference for each day was sorted then plotted. The plot on the right shows the tail distribution starting from the 99th percentile.

5.2.2 Constant Risk Premium

Here, we consider the empirical properties of the risk premium, which we are assuming to be constant in our model. To quantify to what extent the conditional risk premium may change over time, we regress the realized risk premium on the lagged values of both price volatility σ and the log ratio of the trading volume to pool liquidity, $\log(\frac{V}{c})$. The first predictor, recent price volatility, captures the possibility that the riskiness of the spread between returns of liquidity providers and the buy-and-stake return is affected by the price volatility of the risky asset in the pool. The second variable

is a natural predictor of the risk premium: if the pool size were to fall relative to the expected trading volume, liquidity providers would be expected to earn abnormally high returns. Results of the regressions can be found in figure 4.

We find only weak evidence, if any, that the risk premium is predictable by the recent price volatility. However, there is some statistically significant evidence that the volume-to-liquidity ratio can forecast the conditional risk premium. This may reflect adjustment frictions faced by the liquidity providers, which we do not model explicitly. If imperfect pool-size adjustment is the cause of the observed predictability, its effect on the predictable component of the risk premium, as revealed by the above specification, is of economically small magnitude and tends to dissipate after approximately two weeks. While adjustment frictions could be incorporated into the model, we choose not to do that in the basic version of the model presented here.

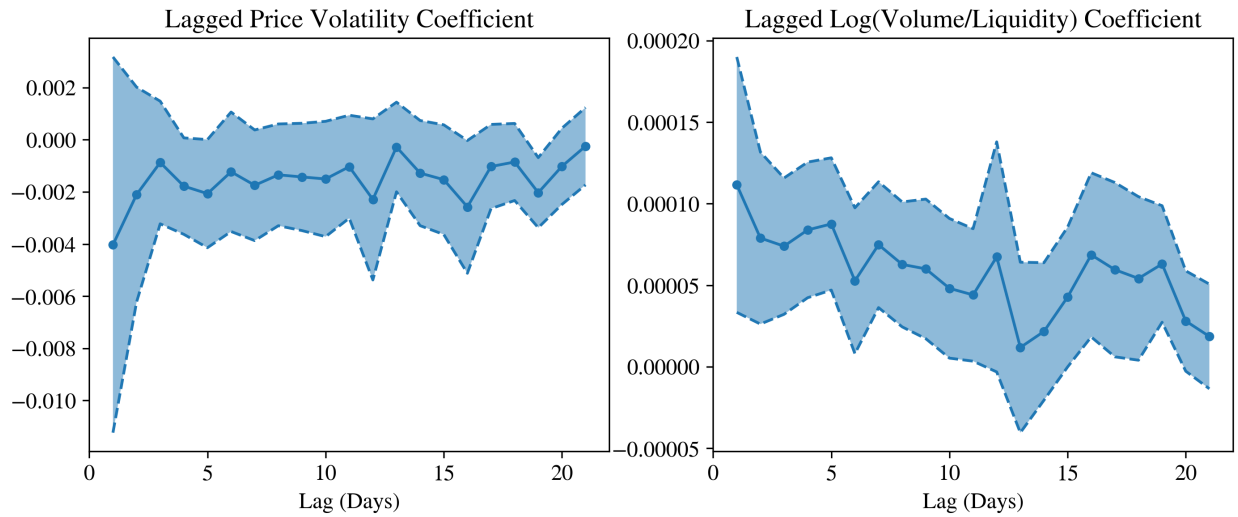

Figure 4: Coefficient of lagged σ and $\log(V/c)$ fit on the spread of the break-even condition with varying lags. The shaded regions are the 95% asymptotic confidence bands around the regression coefficients.

5.2.3 Price Spread Effect

Here we show that impermanent loss is not predictable with respect to price spread, backing our modeling assumption that the pool price matches the external price in expectation at the end of the period. First, we compute the spread Δ between the pool price and external price at the start of each day along with the realized impermanent loss with respect to BNS. We define the spread as the percentage difference between the pool price and external price with respect to the external price. The calculation for realized impermanent loss is a bit more involved. We use LP token data and pool reserve amounts for the Uniswap V2 ETH-USDC pool to determine LP token values at

the start and end of each day. However, pool reserve amounts include fees earned, so in order to determine the value of an LP's initial investment, we deduct these fees from the pool reserve amounts at the end of each day. Finally, we compare the returns of a BNS portfolio against the value of the pool reserve less fees at the end of each day to determine impermanent loss.

Concretely, impermanent loss for each day is computed using the following equations. R represents pool reserve amounts, TS represents LP token supply, F is the fees collected for each asset throughout the day, and γ represents the staked return rate for that particular day. Times 0 and 1 denote start and end of day.

$$TV_0 = \left(R_0^{USDC} + R_0^{ETH} \cdot p_0\right) / TS_0 \tag{25}$$

$$TV_1 = \left(R_1^{USDC} - F^{USDC} + \left(R_1^{ETH} - F^{ETH}\right) \cdot p_1\right) / TS_1 \tag{26}$$

$$TV_1^{BNS} = \left(R_0^{USDC} \cdot \gamma_{USDC} + R_0^{ETH} \cdot \gamma_{ETH} \cdot p_1\right) / TS_0 \tag{27}$$

$$\mathcal{L} = \frac{TV_1^{BNS} - TV_1}{TV_0} \tag{28}$$

We then ran linear regressions using OLS to determine the predictability of impermanent loss with respect to spread at start of day and lagged price volatility. Results are summarized in table 2. We can clearly see that price spread at start of day does not predict impermanent loss given the high standard errors and p-values.

Parameter	**Coefficient**	**Std Err**	**P-Value**
const	0.0002	1.29e-05	0
Δ	-0.0041	0.006	0.483

Parameter	**Coefficient**	**Std Err**	**P-Value**
const	0.0001	6.34e-05	0.090
Δ	-0.0036	0.006	0.532
Lagged σ	0.0045	0.003	0.086

Table 2: Linear regression parameters predicting impermanent loss. Two regressions were run, one just using spread and one with spread and lagged price volatility σ.

5.2.4 Break-even Condition

To show that buy-and-stake (BNS) is a better baseline than the traditionally considered buy-and-hold (BNH) portfolio, we compare each to the real-world returns LPs make on the ETH-USDC Uniswap V2 pool. We use data from AAVE and Lido to determine the staked returns for both ETH and USDC and compare the spread between the BNS/BNH and LP returns. Our findings are summarized in table 3. Overall, we find that BNS is significantly better at tracking the returns of LPs.

BNS Spread	BNH Spread	|BNS Spread|	|BNH Spread|	Window
$-8.508 \cdot 10^{-6}$	$9.5427 \cdot 10^{-5}$	$1.3617 \cdot 10^{-4}$	$1.8283 \cdot 10^{-4}$	1 day

Table 3: Spreads for BNH and BNS returns compared to LP returns. The first two columns are the mean spread for each portfolio, and the third and fourth columns are the mean of the absolute value of the spreads.

5.3 Parameter Estimation

To estimate parameters, we apply the generalized method of moments (GMM). We aim to estimate four parameters in total: a_0, a_c, a_σ, and RP.

We define unconditional GMM moments by conditioning down the volume equation, (22), and the conditional break-even condition for the LPs, (12). We list the resulting moment conditions in (29–42) below.

The moment $\mathbb{E}[g_0] = 0$ indicates that the residual ε_t in the volume equation has zero mean. Conditions $\mathbb{E}[g_i] = 0$, $i = 3, 4, 5$ state that the residual in the volume equation is not correlated with lagged values of price volatility, trade count, and trade size.

The moment $\mathbb{E}[g_1] = 0$ is the average break-even condition, stating that the average spread between the LP return implemented by the model and the empirical buy-and-stake return is equal to the risk premium. Other unconditional versions of the break-even condition, expressed as $\mathbb{E}[g_i] = 0$, $i = 6, 7, 8, 9, 10$, are obtained by interacting the LP vs. buy-and-stake return spread with the lagged values of price volatility, log liquidity, log trading volume, trade count, and trade size. As we discussed above, the size of the pool relative to the predictors of trading volume is a natural candidate for predicting the return spread earned by liquidity providers.

The moment $\mathbb{E}[g_2] = 0$ states that the average spread between the empirical LP return and the buy-and-stake return is equal to the risk premium – this moment identifies the risk premium

parameter RP. Moments $\mathbb{E}[g_i] = 0$, $i = 11, 12, 13$ require the risk premium to be uncorrelated with the lagged values of the log pool size, log trading volume, and price volatility.

In summary, we define our moments as

$$g_0(t) = V_t - (\bar{a}_0 + a_\sigma \cdot \sigma_{t-1} + a_c \cdot \log c_{t-1}) \tag{29}$$

$$g_1(t) = f \cdot (\bar{a}_0 + a_\sigma \cdot \sigma_{t-1} + a_c \cdot \log c_{t-1}) \cdot \frac{1}{c_{t-1}} + \sqrt{v_t} - BNS_t - RP \tag{30}$$

$$g_2(t) = LP_t - BNS_t - RP \tag{31}$$

$$g_3(t) = g_0(t) \cdot \sigma_{t-1} \tag{32}$$

$$g_4(t) = g_0(t) \cdot TC_{t-1} \tag{33}$$

$$g_5(t) = g_0(t) \cdot TS_{t-1} \tag{34}$$

$$g_6(t) = g_1(t) \cdot \sigma_{t-1} \tag{35}$$

$$g_7(t) = g_1(t) \cdot \log(c_{t-1}) \tag{36}$$

$$g_8(t) = g_1(t) \cdot \log(V_{t-1}) \tag{37}$$

$$g_9(t) = g_1(t) \cdot TC_{t-1} \tag{38}$$

$$g_{10}(t) = g_1(t) \cdot TS_{t-1} \tag{39}$$

$$g_{11}(t) = g_2(t) \cdot \log(c_{t-1}) \tag{40}$$

$$g_{12}(t) = g_2(t) \cdot \log(V_{t-1}) \tag{41}$$

$$g_{13}(t) = g_2(t) \cdot \sigma_{t-1} \tag{42}$$

$$\tag{43}$$

where v_t is the external price movement during time window t, V is the real noise volume, BNS is the baseline BNS portfolio based on real data, and LP is real LP returns assuming entry and exit at the beginning and end of the time window.

With our moments defined, we run a two-step GMM (e.g., Cochrane (2009)). Our GMM is run on daily windows of our data, resulting in over 800 data points. We scale both liquidity c and trading volume c to be in units of \$1M USD for more reasonable parameter values. We use the identity weight matrix for the first step. In the second step, we specify the weight matrix as the inverse of the diagonal matrix of the moment variances (using the estimated parameters from the first step to evaluate the moments). We find little change in estimated parameter value between the two steps.

We summarize parameter estimates and their standard errors in Table 4. An example of the

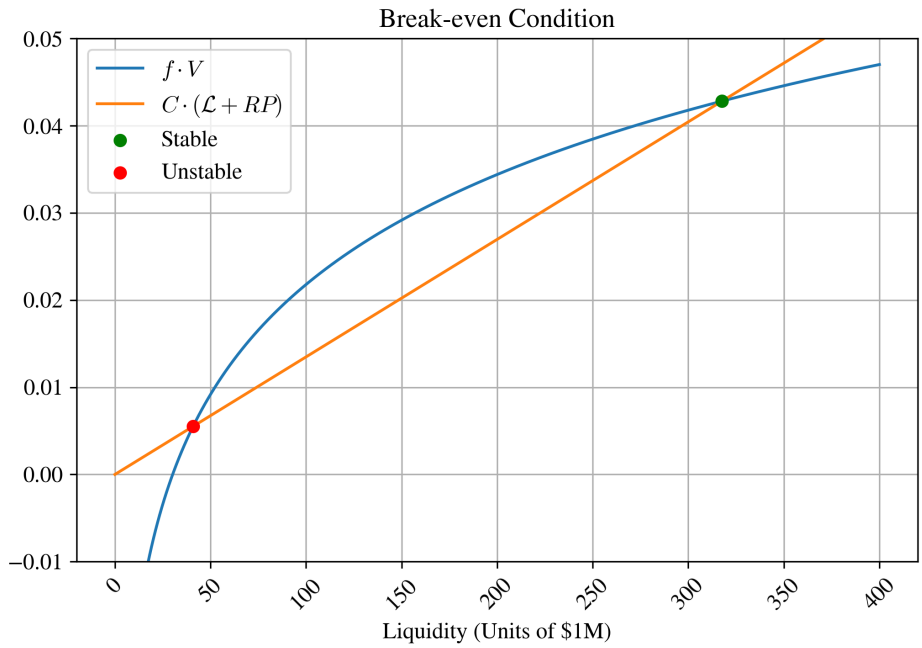

Figure 5: A visual depiction of the break-even condition using parameters fit from GMM. $\mathcal{L} = 0.000135$ and $\sigma = 0.467$ were chosen from the medians of our data and a fee of 0.002 was used.

break-even condition with our fitted parameters can be seen in Figure 5.

Parameter	**Value**	**Std Err**	**95% Confidence Interval**
a_0	-21.6456	6.0736	-33.5499, -9.7412
a_c	5.1079	1.4866	2.1942, 8.0216
a_σ	11.1560	1.6743	7.8743, 14.4376
RP	-1.7941e-5	9.7519e-6	-3.7055e-5, 1.1724e-6

Table 4: Estimated values of our model parameters using Generalized Method of Moments. The standard error was calculated using the Newey-West approxmiation for the spectral density matrix with $k = 6$. The liquidity and trading volume values used in GMM were scaled down by 10^6 to achieve more reasonable values.

5.3.1 Relation between Optimal Fee and σ

Having estimated model parameters, we explore the relationship between optimal fee and price volatility σ. We assume a log normal distribution for the percent change in price with parameters μ, σ, which allows us to characterize the expected impermanent loss as

$$\mathbb{E}[\mathcal{L}] = \gamma\left(\exp\left(\mu + \frac{\sigma^2}{2}\right)\right) - \exp\left(\frac{\mu}{2} + \frac{\sigma^2}{8}\right) \tag{44}$$

We then compute the optimal fee over varying σ and plot the results in Figure 6. The range of a_f used is identified from the estimation exercise outlined in section 6.2. This can be thought of as the optimal fee the platform should set as a function of their belief of the next period's price volatility σ. As shown, there is a clear positive relation between the optimal fee and price volatility. As price becomes more volatile, fee should increase. This result motivates the need for dynamic fees, which we explore in the following section.

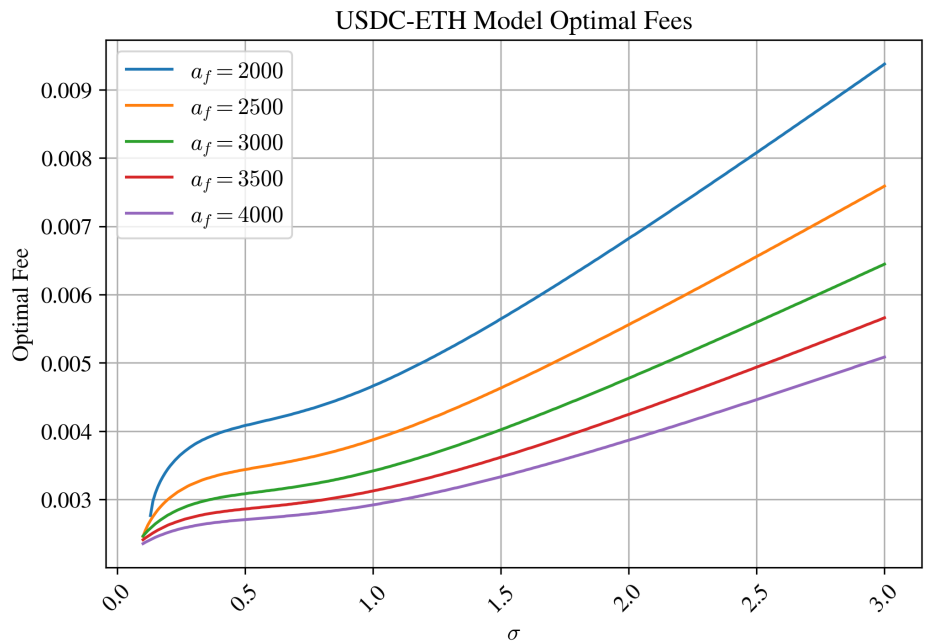

Figure 6: Daily optimal fee f^* with varying annualized σ and $\mu = 0$ to represent an equal chance of the price moving up or down. The values used for staked USDC and ETH returns were the average over the data collected 0.98 and 1.1 basis points, respectively.

6 Dynamic Fees

In this section we explore the impact of a dynamic AMM fee in a simulated environment, calibrated to Uniswap v2 data. We find economically significant differences between the dynamic fee process implied by our model and the current constant fee structure in Uniswap v2.

To implement the dynamic fee policy, two additional elements are needed: a predictive model of impermanent loss $\mathcal{L}$ (which depends on future prices and volatility), and the parameter a_f,

describing the sensitivity of the trading volume to transasction fees.[7] While we cannot identify a_f empirically, we define below an implied range for a_f values based on the Uniswap v2 data, and perform sensitivity analysis with respect to a_f over the empirically relevant range.

This section is organized as follows: first, we describe the simulation of the trading model. Then we determine the relevant implied range for a_f. Next, we analyze the relation between the dynamic fees and price volatility and compare the performance of the dynamic fee policy to the static fee used by Uniswap v2.

6.1 Simulation

We simulate the decision process of both the platform and the LPs. For a given day t, the platform first computes the optimal fee assuming that the LPs strategically deposit liquidity according to (20), conditionally on σ_{t-1}. Liquidity providers act next. We then compute the expected trade volume given σ_{t-1}; the price moves according to the real price process from Binance; and profits are realized at the end of the day.

To define the break-even liquidity, we need to model expectations of impermanent loss by the platform operators and by liquidity providers. For our simulation experiments, we make a simplifying assumption that both groups use recent price volatility as the only predictor of impermanent loss, and that their conditional expectation of impermanent loss is described by a three-parameter (L, k, σ_0) logistic function on lagged price volatility σ.

$$\mathbb{E}[\mathcal{L}_t|\sigma_{t-1}] = \frac{L}{1 + \exp(-k(\sigma_{t-1} - \sigma_0))} \tag{45}$$

Figure 7 shows the fit of our model of the expected impermanent loss. The choice of the statistical model for the expected impermanent loss affects the precise response of the optimal fee to changes in conditional price volatility. However, the comparison of the fixed and dynamic fee rules should be less affected, because we evaluate both rules using the same forecast of $\mathcal{L}$.

Finally, we use estimated parameters a_0, a_c, a_σ on data from Uniswap V2 between September 2022 and December 2024 and run the simulation on price data from Binance. We use the same time window of one day as our GMM estimation. For each window t, we compute the annualized σ_t from the one-second granular price data. For each time window, we compute optimal liquidity,

[7] The sensitivity parameter a_f cannot be identified with the constant fee history under Uniswap v2. Uniswap v3 data provides an avenue, however mapping v3 to v2 proves to be a non-trivial task. In the future, it should be feasible to estimate a_f under the Uniswap v4 protocol.

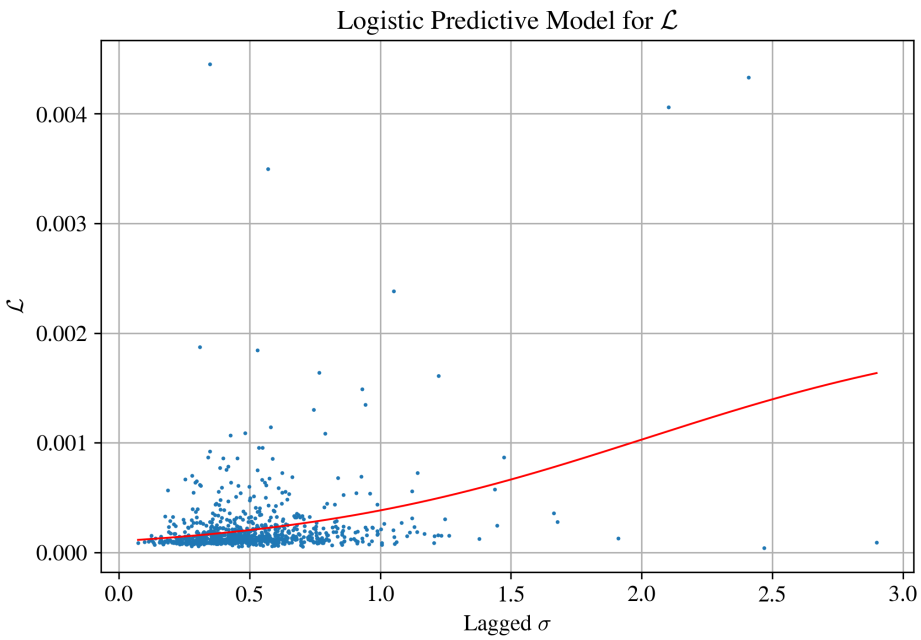

Figure 7: A fit of the logistic predictive model for impermanent loss $\mathcal{L}$ as a function of lagged price volatility σ. The fitted parameters used were $L = 0.0021, k = 27.7716, \sigma_0 = 0.1064$.

fee, and expected trading volume for both the fixed and dynamic fee policies.

6.2 Implied range of a_f

As mentioned, estimation opportunities for fee sensitivity a_f are limited until v4 experiments can be implemented. Nonetheless, we can make some progress by assuming that the current 0.3% fee is equal to the average value of the optimal dynamic fee over our sample period. We run a simulation of daily dynamic fees for the same dates as the GMM parameters were fit on (Sept 2022 - Dec 2024). To find the optimal fee, for each day t, we iterate over all fees between $0-3\%$ with a step of 0.00001 (1/10 of a bp) and compute the break-even liquidity in response to fee and σ_{t-1}. We then pick the fee that results in the maximal break-even liquidity and compute trading volume as a function of σ_t. This can be thought of as the process the platform goes through to determine the optimal fee to set, in each period.

We find that a range of roughly $2500 \leq a_f \leq 3500$ is "reasonable", with $a_f \approx 3250$ being the closest to having an average fee of the current status quo of 0.003. As expected, as a_f increases, the optimal fee declines.

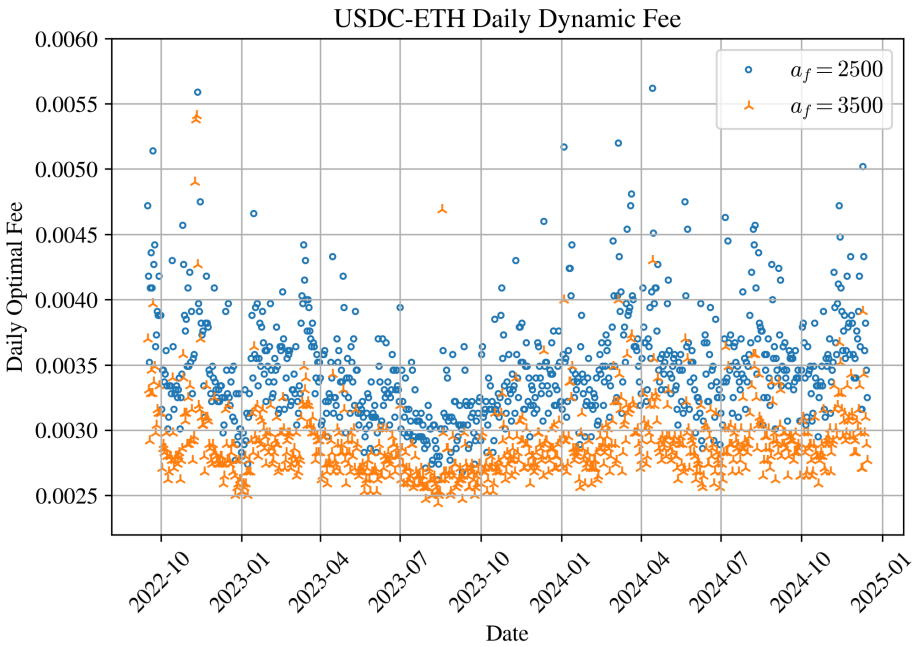

Figure 8: Optimal daily dynamic fees for price data from September 2022 to December 2024 for $a_f = 2500, 3500$.

6.3 Impact of Dynamic Fees

We quantify the benefits of dynamic fee AMMs relative to fixed fee AMMs using a_f values within the range determined from our previous analysis. We employ the same logistic model to forecast impermanent loss and utilize out-of-sample price data from December 17, 2024 to May 9, 2025 (144 days).

For each trading day, we simulate aggregate fees earned under dynamic versus fixed fee structures. Dynamic fees follow the process described above, while fixed fees assume LPs optimize their response to a constant 0.3% fee rate. Solutions were found on all days for both dynamic and static fee mechanisms. Consistent with our earlier methodology, we exclude these days from the analysis.

Figure 9 presents the cumulative fee differential, while Table 5 reports average liquidity increases under dynamic pricing. Using our estimated $a_f = 3250$, the dynamic mechanism generates approximately \$440,000 in additional fees annually—an 13.2% annual improvement over fixed fees. Moreover, average liquidity increases by 2.6%.

These results demonstrate that dynamic fee AMMs achieve superior efficiency relative to fixed fee alternatives, even accounting for forecasting errors inherent in practical implementation (as shown in Figure 7).

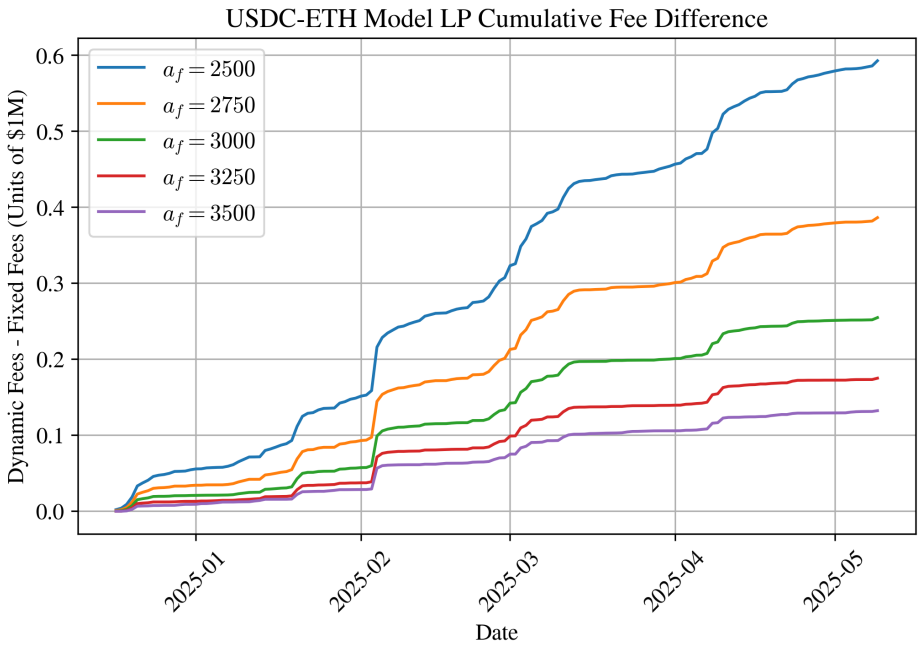

Figure 9: The cumulative difference in fees collected by the platform with dynamic fees vs. a fixed 0.003 fee. Units of fees are in $1,000,000. In ascending order, each a_f resulted in 43.7%, 29.7%, 20.0%, 13.2%, and 8.8% additional cumulative fees over a fixed fee, annualized.

a_f	2500	2750	3000	3250	3500
Average Liquidity Increase	9.9%	6.1%	3.8%	2.6%	2.0%

Table 5: The additional percentage of liquidity a dynamic fee AMM generates versus a fixed 0.003 fee AMM. The percent increase $(D - F)/F$ where D is dynamic liquidity and F is fixed liquidity was computed for each day between Dec 17 2024 - May 9 2025.

7 Discussion

We define a structural model of AMMs to determine how and when fees should be dynamically adjusted, addressing a question central to classical market microstructure but largely unexplored in decentralized markets. Unlike traditional markets, which use volatility-sensitive spreads to mitigate adverse selection, AMMs currently rely on static fees. Our model captures the interconnected decisions of platforms, liquidity providers, and traders to describe the behavior of a "stable-risk" asset pair AMM. Using a three-period game, we analyze how agents jointly determine the AMM fee, trade volume, and liquidity provisioned. We derive equilibrium conditions for both the optimal fee and liquidity, establishing a positive relationship between price volatility and the optimal fee.

Our key contribution lies in combining a rigorous theoretical framework with empirical struc-

tural estimation that addresses endogeneity between trading volume and liquidity provision, a gap in prior research. Using a two-stage generalized method of moments (GMM) approach, we provide robust empirical evidence that optimal AMM fees should dynamically adjust to asset price volatility. This contribution bridges traditional and decentralized finance, revealing that AMMs and traditional markets share common design principles for dynamic fee structures, despite their significant microstructure differences.

Our findings show that static fees in AMMs lead to inefficiencies, such as lower liquidity and reduced trading volumes, underscoring the critical need for dynamic fee mechanisms. While the absence of CFMMs with dynamic fees limits full quantification of the volatility-fee relationship, our results offer actionable insights for designing more efficient and scalable AMMs.

Future research could refine and extend our model in several ways. For example, while our functional form for trading volume empirically captures its high-level relationship with fees, liquidity, and volatility, it does not fully account for edge conditions, such as trading volume approaching zero when fees approach one or liquidity falls to zero. Uniswap v3's multiple fee tiers provide a potential data source, though mapping this to an equivalent CFMM model is challenging due to bonding constraints. Alternatively, Uniswap v4 offers an opportunity to design and test a dynamic fee AMM, enabling further validation of our findings. These extensions could deepen our understanding of trading behavior under dynamic fees, advancing both theory and practice in decentralized markets.

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Appendix

A Proofs

Proof of lemma 1. Intuitively, since BNS is strictly better than a buy-and-hold portfolio, it must be the case that this is true. More formally, we see

$$\gamma(\mathbb{E}[v]) - \mathbb{E}[\sqrt{v}] \geq 0 \tag{46}$$

$$\frac{e^{\gamma_s \cdot dt} + e^{\gamma_r \cdot dt} \cdot \mathbb{E}[v]}{2} - \mathbb{E}[\sqrt{v}] \geq 0 \tag{47}$$

Note that by definition we have that the staked returns $\gamma_s, \gamma_r \geq 1$. Thus,

$$\frac{e^{\gamma_s \cdot dt} + e^{\gamma_r \cdot dt} \cdot \mathbb{E}[v]}{2} - \mathbb{E}[\sqrt{v}] \geq \frac{1 + \mathbb{E}[v]}{2} - \mathbb{E}[\sqrt{v}] \geq 0 \tag{48}$$

Let $\varphi(v)$ be the PDF for the distribution of v. We then see

$$\int \varphi(v) \left(\frac{1+v}{2} - \sqrt{v} \right) dv \geq 0 \tag{49}$$

Since v only takes on non-negative values, by AM-GM this must always be true. □

Proof of lemma 2. Consider the derivative of the break-even with respect to f. Note that c is a function of f.

$$f \cdot V(f, \sigma, c) = c \cdot (\mathcal{L} + RP) \tag{50}$$

$$V(f, \sigma, c) + f \cdot \left(\frac{\partial V}{\partial f} + \frac{\partial V}{\partial c} \cdot \frac{\partial c}{\partial f} \right) = \frac{\partial c}{\partial f} \cdot (\mathcal{L} + RP) \tag{51}$$

We assume σ is exogenously determined, so it is not dependent on f. By definition, $\frac{\partial c}{\partial f} = 0$ when the platform picks the best fee f^* such that liquidity is maximized. Thus, the first-order condition of the break-even states that

$$V(f, \sigma, c) + f \cdot \frac{\partial V}{\partial f} = 0 \tag{52}$$

Finally, since $V, f > 0$, it must be the case that $\frac{\partial V}{\partial f} < 0$. □

Proof of theorem 1. Taking the first order condition with respect to f of the break-even results in the following equation:

$$\frac{\partial}{\partial f}c = \frac{a_0 - 2a_f \cdot f + a_\sigma \cdot \sigma}{\mathcal{L} + RP} = 0 \tag{53}$$

$$a_0 - 2a_f \cdot f + a_\sigma \cdot \sigma = 0 \tag{54}$$

$$f = \frac{1}{2a_f}\left(a_0 + a_\sigma \cdot \sigma\right) \tag{55}$$

□

Proof of theorem 2. Manipulating the break-even, we see that

$$f \cdot V(f, \sigma, c) = c \cdot (\mathcal{L} + RP) \tag{56}$$

$$f \cdot (a_0 - a_f \cdot f + a_\sigma \cdot \sigma + a_c \cdot \log(c)) = c \cdot (\mathcal{L} + RP) \tag{57}$$

$$\frac{c \cdot (\mathcal{L} + RP) - f \cdot (a_0 - a_f \cdot f + a_\sigma \cdot \sigma)}{f \cdot a_c} = \log(c) \tag{58}$$

$$\exp\left(\frac{c \cdot (\mathcal{L} + RP) - f \cdot (a_0 - a_f \cdot f + a_\sigma \cdot \sigma)}{f \cdot a_c}\right) = c \tag{59}$$

$$\exp\left(\frac{c \cdot (\mathcal{L} + RP)}{f \cdot a_c}\right) \cdot \exp\left(\frac{-f \cdot (a_0 - a_f \cdot f + a_\sigma \cdot \sigma)}{f \cdot a_c}\right) = c \tag{60}$$

Let $u = \frac{c \cdot (\mathcal{L}+RP)}{f \cdot a_c}$, so $c = \frac{u \cdot f \cdot a_c}{\mathcal{L}+RP}$.

$$\frac{u \cdot f \cdot a_c}{\mathcal{L} + RP} = \exp(u) \cdot \exp\left(\frac{a_f \cdot f - a_0 - a_\sigma \cdot \sigma}{a_c}\right) \tag{61}$$

$$u \cdot \exp(-u) = \frac{\mathcal{L} + RP}{f \cdot a_c} \cdot \exp\left(\frac{a_f \cdot f - a_0 - a_\sigma \cdot \sigma}{a_c}\right) \tag{62}$$

$$-u = W\left(-\frac{\mathcal{L} + RP}{f \cdot a_c} \cdot \exp\left(\frac{a_f \cdot f - a_0 - a_\sigma \cdot \sigma}{a_c}\right)\right) \tag{63}$$

Finally, substituting c back in for u, we have

$$-\frac{c \cdot (\mathcal{L} + RP)}{f \cdot a_c} = W\left(-\frac{\mathcal{L} + RP}{f \cdot a_c} \cdot \exp\left(\frac{a_f \cdot f - a_0 - a_\sigma \cdot \sigma}{a_c}\right)\right) \tag{64}$$

$$c^* = -\frac{f \cdot a_c}{\mathcal{L} + RP} \cdot W\left(-\frac{\mathcal{L} + RP}{f \cdot a_c} \cdot \exp\left(\frac{a_f \cdot f - a_0 - a_\sigma \cdot \sigma}{a_c}\right)\right) \tag{65}$$

□

Proof of theorem 3. Consider the rate of change with respect to c of both sides of the break-even

constraint. On the RHS, by assumption, it is simply a positive constant $\mathcal{L} + RP$. On the LHS, we have $\frac{a_c}{c}$. Note that as $c \to \infty$, the rate at which it changes approaches 0. Thus, if there are solutions to the break-even, it must be the case that the largest solution crosses the RHS from top to bottom, meaning it is stable.

□

B Arbitrageur Behavior

Here, we describe the behavior of arbitrageurs. We assume that arbitrageurs incur AMM trading costs (including fees and slippage), but are otherwise able to frictionlessly trade between the AMM and an external market to realize the arbitrage. They trade the quantity that maximizes their profit after the price shock v has been realized. Note that if $v > 1$, a potential profit opportunity exists by buying from the AMM and selling externally. Similarly, if $v < 1$, a potential profit opportunity is to buy externally and sell to the AMM.

We assume initial reserve amounts of ℓ_s, ℓ_r, which implies $p_0 = \frac{\ell_s}{\ell_r}$. Their optimization problem depends on the realized value of the price multiplier v, and is given by

$$\max_{x \geq 0} \begin{bmatrix} x \cdot v \cdot p_0 - x \cdot p_e^{buy}(c, f, x) & \text{if } v \geq 1 \\ \\ x \cdot p_e^{sell}(c, f, x) - x \cdot v \cdot p_0 & \text{if } v < 1. \end{bmatrix} \tag{66}$$

Case 1: $v > 1$, i.e., the arbitrageur buys from the AMM and sells externally for profit.

$$g(x) \stackrel{\text{def}}{=} \max_{0 \leq x < \ell_r} v \cdot \frac{\ell_s}{\ell_r} \cdot x - \frac{1}{1-f}\left(\frac{\ell_s \ell_r}{\ell_r - x} - \ell_s\right) \tag{67}$$

Then

$$\frac{dg}{dx} = v \cdot \frac{\ell_s}{\ell_r} - \frac{\ell_s \ell_r}{(1-f)(\ell_r - x)^2} \tag{68}$$

$$\frac{d^2 g}{dx^2} = -\frac{2\ell_s \ell_r}{(1-f)(\ell_r - x)^3} \tag{69}$$

Note that $\frac{d^2 g}{dx^2} \leq 0$ for $0 \leq x < \ell_r$. This means that if $\frac{dg}{dx}(0) < 0$, then $\frac{dg}{dx} < 0$ for all x.

$$\frac{dg}{dx}(0) = v \cdot \frac{\ell_s}{\ell_r} - \frac{\ell_s}{(1-f)\ell_r} \tag{70}$$

So if $\frac{1}{1-f} > v > 1$, we have that the optimal x is $x = 0$. We can also check that

$$v < \frac{1}{1-f} \Rightarrow v \cdot \frac{\ell_s}{\ell_r} \cdot x - \frac{1}{1-f}\left(\frac{\ell_s \ell_r}{\ell_r - x} - \ell_s\right) \leq 0 \text{ for all } x > 0 \tag{71}$$

So for the rest of the analysis, we assume $v > \frac{1}{1-f}$ In this case, the first-order condition (given by Equation 68) becomes

$$v = \frac{\ell_r^2}{(1-f)(\ell_r - x)^2} \tag{72}$$

Solving for x, we have

$$x = \ell_r \pm \frac{\ell_r}{\sqrt{(1-f)v}} \tag{73}$$

Since we assume $x \leq \ell_r$, we have

$$x = \ell_r - \frac{\ell_r}{\sqrt{(1-f)v}} \tag{74}$$

Thus the informed trader's best response is

$$x = \begin{cases} 0 & \text{if } 1 < v \leq \frac{1}{1-f} \\ \ell_r - \frac{\ell_r}{\sqrt{(1-f)v}} & \text{if } v > \frac{1}{1-f} \end{cases} \tag{75}$$

Case 2: When $v < 1$, then the arbitrageur wishes to buy externally and sell to the AMM.

$$g(x) \stackrel{\text{def}}{=} \max_x \ \ell_s - \frac{\ell_s \ell_r}{\ell_r + (1-f)x} - x \cdot v \cdot \frac{\ell_s}{\ell_r} \tag{76}$$

Then

$$\frac{dg}{dx} = \frac{(1-f)\ell_s \ell_r}{(\ell_r + (1-f)x)^2} - v \cdot \frac{\ell_s}{\ell_r} \tag{77}$$

$$\frac{d^2 g}{dx^2} = -\frac{2(1-f)^2 \ell_s \ell_r}{(\ell_r + (1-f)x)^3} \tag{78}$$

The first-order condition becomes

$$v = \frac{(1-f)\ell_r^2}{(\ell_r + (1-f)x)^2} \tag{79}$$

Solving for x, we have

$$x = \frac{\ell_r}{\sqrt{(1-f)v}} - \frac{\ell_r}{1-f} \tag{80}$$

Note that x is positive only if $v < 1 - f$. If $v > 1 - f$, the optimization (Equation 76)

$$\ell_s - \frac{\ell_s \ell_r}{\ell_r + (1-f)x} - v \cdot \frac{\ell_s}{\ell_r} \cdot x < 0 \text{ for all } x \neq 0 \tag{81}$$

So the optimal x in this case is $x = 0$

$$x = \begin{cases} 0 & \text{if } 1 > v > 1 - f \\ \\ \frac{\ell_r}{1-f} - \frac{\ell_r}{\sqrt{(1-f)v}} & \text{if } v < 1 - f \end{cases} \tag{82}$$

Thus, the arbitrageur's optimal response for any price change v is

$$x = \begin{cases} 0 & \text{if } 1 - f \leq v \leq \frac{1}{1-f} \\ \\ \ell_r - \frac{\ell_r}{\sqrt{(1-f)v}} & \text{if } v > \frac{1}{1-f} \\ \\ \frac{\ell_r}{1-f} - \frac{\ell_r}{\sqrt{(1-f)v}} & \text{if } v < 1 - f \end{cases} \tag{83}$$